

Customer Resolution Agent — Architecture

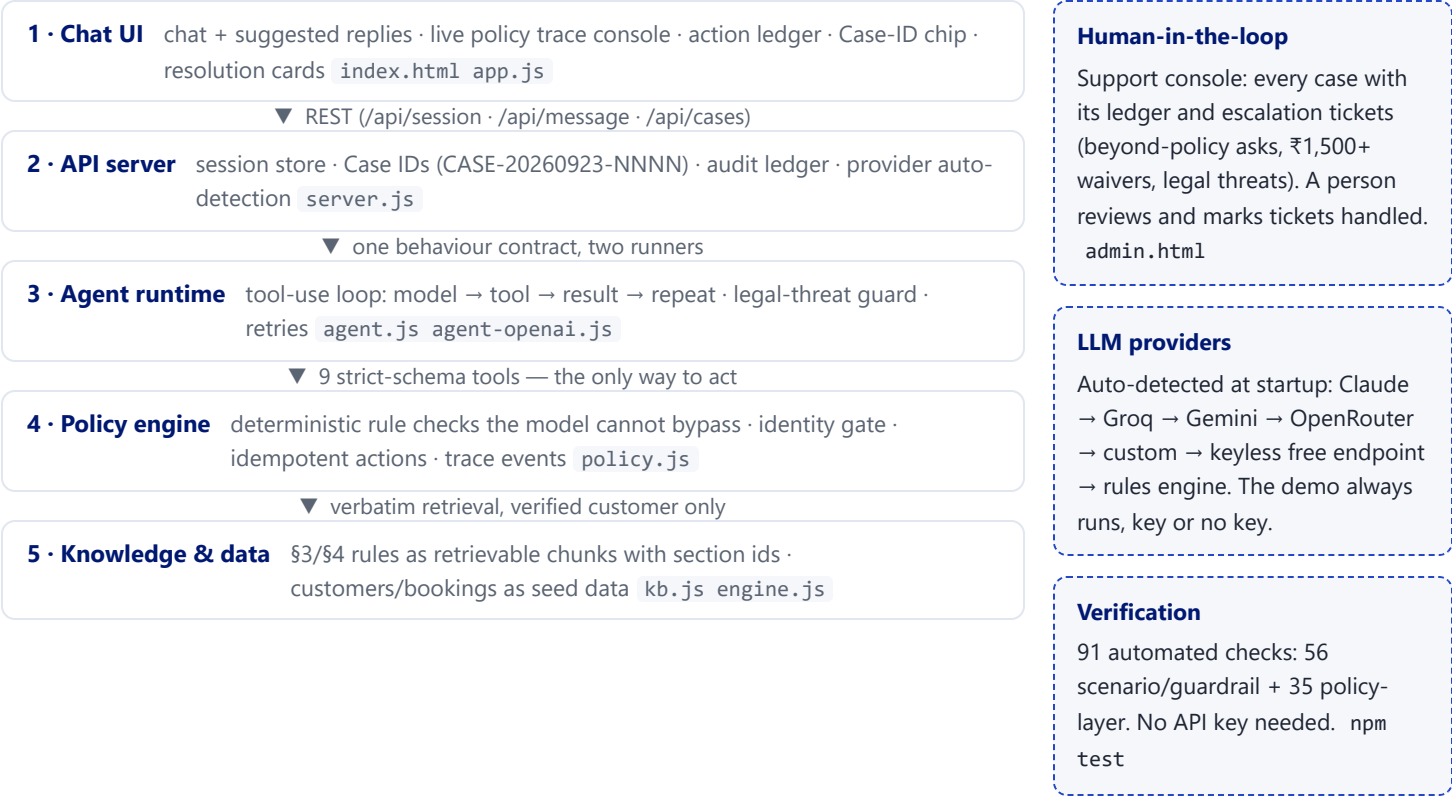
Airline disruption resolution (AIONOS Assignment 3) · grounded only in the supplied data pack · Node.js, zero-dependency core

CORE PRINCIPLE



The model never decides entitlements. It requests actions; deterministic code computes eligibility from the data pack and either executes or returns the blocking rule, which the agent must relay and escalate.

SYSTEM LAYERS



THE NINE TOOLS (POLICY-ENFORCED)

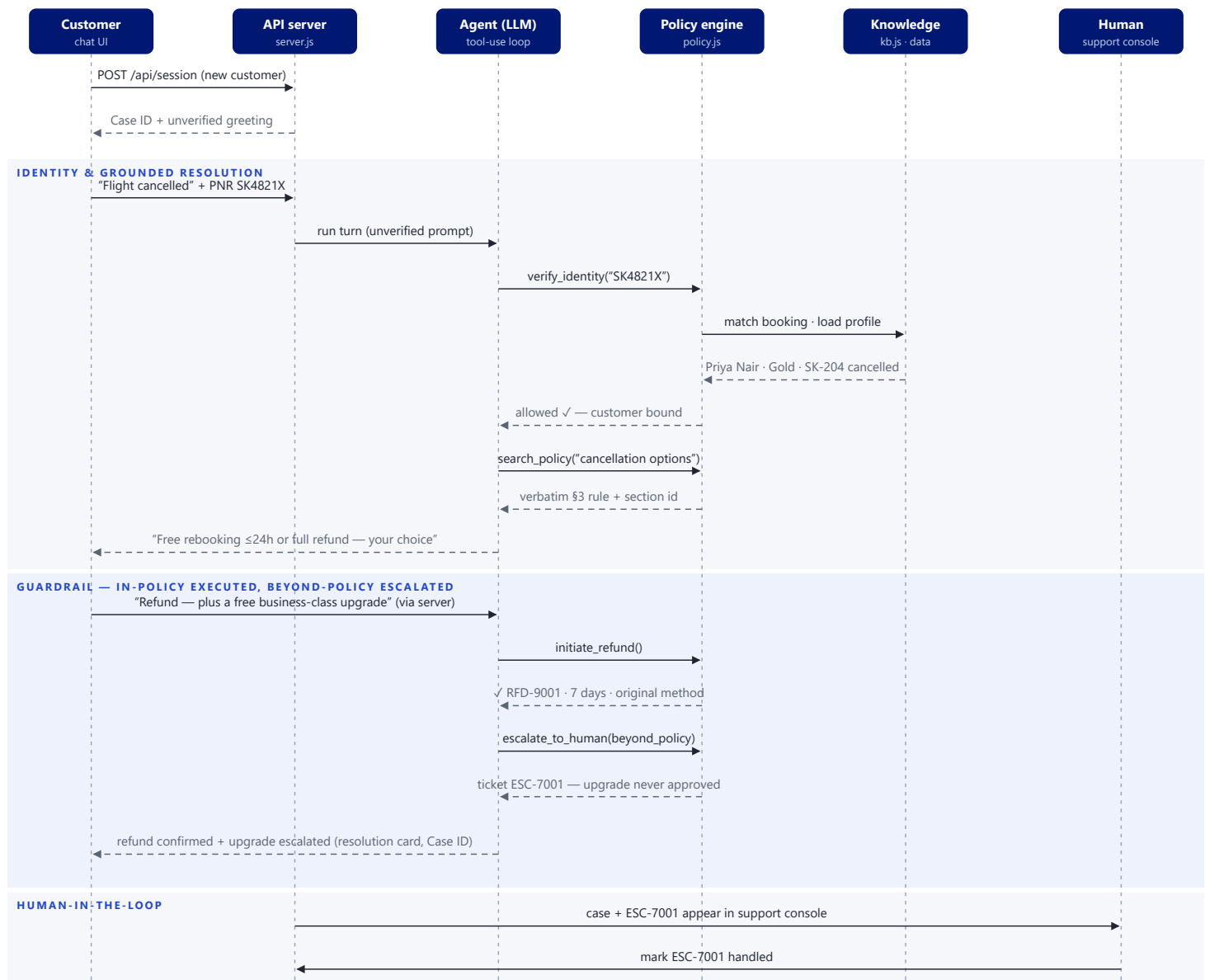
`verify_identity` `search_policy` `get_booking_details` · `issue_delay_compensation` `rebook_free` `initiate_refund`
`quote_fare_difference` `rebook_paid_alternative` · `escalate_to_human`

RULES ENFORCED IN CODE (FROM THE DATA PACK)

- **Delay tiers:** <3h → ₹500 meal voucher · >3h → + lounge access · >5h → + hotel for the *delayed hours only*, never a full night.
- **Cancellation:** customer's choice — free rebooking within 24h (Gold/Platinum priority) or full refund in 7 business days, original payment method only.
- **Fare difference:** voluntary move to a higher fare is payable; above ₹1,500 the agent cannot waive — supervisor escalation (Meher's ₹2,000 case).
- **Loyalty:** priority rebooking only — no extra compensation; free upgrades are beyond policy → escalated, never approved.
- **Hard boundaries:** legal threats escalate immediately; non-airline disruptions get no exceptions; unverified sessions cannot touch any account action.

Sequence — identity → grounded resolution → guardrail → human

Priya's scenario end-to-end: verification, in-policy refund executed, beyond-policy upgrade escalated. Solid = call · dashed = response.



Every arrow into the policy engine emits an audit event: policy trace (Allowed / Blocked / Supervisor / Escalated), ledger entry, or ticket — all tied to the Case ID.

IF YOU HAVE TO EXPLAIN IT IN THREE SENTENCES

- **The LLM never decides entitlements** — it requests tools; the policy engine computes eligibility from the data pack and either executes or returns the blocking rule (a hotel on a 4-hour delay is refused *by code*, with the ">5 hours" rule attached).
- **Everything is auditable** — each turn emits a policy trace, ledger ids (VCH/LNG/HTL/RFD/RBK) and tickets (ESC), tied to a Case ID and visible live in both consoles.
- **Escalation is the boundary, not a failure** — beyond-policy asks, waivers above ₹1,500 and legal threats route to a human immediately; the agent never approves outside policy and never promises an outcome.